Introduction to Quantitative Biology
Lecture Notes: Excitability, Thresholds, and Transient Cell States

From An Introduction to Systems Biology — Uri Alon

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Excitability in the Competence Control Circuit of *Bacillus subtilis*

Central Question

Certain types of cellular differentiation are probabilistic and transient. In such systems individual cells can switch to an alternative state and, after some time, switch back again. This behaviour can be understood in terms of **excitability** in the underlying genetic circuit.

In *Bacillus subtilis*, competence is an example of such a transiently differentiated state. Upon encountering nutrient limitation, most of *B. subtilis* cells commit irreversibly to sporulation, while a minority of cells become competent for DNA uptake.

How do the genes and proteins interact dynamically in individual cells to control both spontaneous entry into competence and return to vegetative growth?

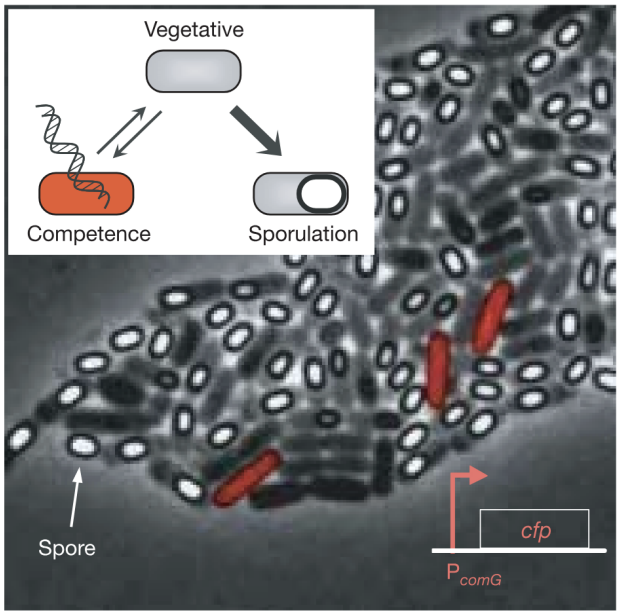


Figure 1. Stress response of a *B. subtilis* microcolony in nutrient-limited conditions. Inset: a flow chart illustrating developmental paths connecting the vegetative state, sporulation and competent state.

In this lecture, we take the 2006 *Nature* paper *An excitable gene regulatory circuit induces transient cellular differentiation* by Süel *et al.* [1] as a case study. The authors developed a mathematical model, and found that a core module containing a fast positive and slow negative feedback exhibits excitability, where relatively small, threshold-crossing perturbations trigger large-amplitude excursions in phase space that eventually return the system to its initial state. Excitable dynamics driven by noise naturally generate stochastic and transient responses, thereby providing an ideal mechanism for competence regulation.

A fast positive feedback + A slow negative feedback $\longrightarrow$ Excitability

1. Background: The Competence Control Circuit

Extensive work has produced a detailed map of the molecular interactions in the *B. subtilis* competence control circuit. At the heart of this circuit is the **ComK** "master" transcription factor. ComK activates a suite of genes required for competence, including the *comG* operon. It also activates its own expression, and this transcriptional **positive-feedback** loop is critical for the induction of competence.

Upon entry into stationary phase, *comK* is expressed at a basal level, but ComK protein is rapidly degraded by the **MecA complex**, a multiprotein assembly containing the ClpP–ClpC proteases. The **ComS** peptide competitively inhibits ComK degradation by MecA. Expression of ComS therefore favours induction of competence by allowing ComK levels to build up sufficiently to enable full ComK activation by positive autoregulation. Conversely, overexpression of ComK suppresses *comS* expression. This interaction implies an indirect **negative-feedback** loop acting on ComK that may control exit from competence.

The interactions among MecA, ComK and ComS are collectively called the **MeKS module**.

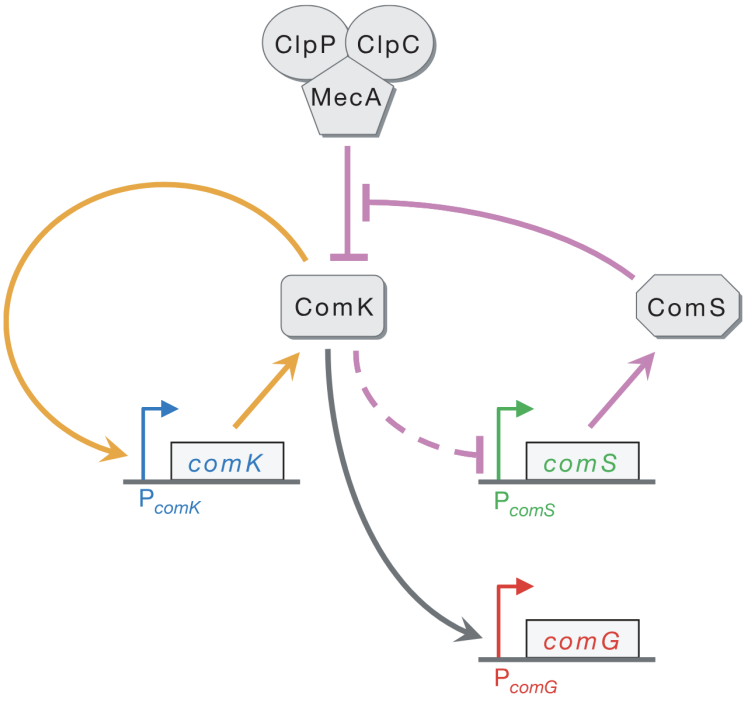


Figure 2. The MeKS module. The transcriptional autoregulatory positive feedback loop of ComK and the ComS-mediated indirect negative feedback loop are depicted in orange and purple, respectively. ComS competes with ComK for degradation by the MecA complex, effectively interfering with degradation of ComK. The dashed purple line from ComK to *comS* denotes indirect repression.

The interaction map of this core competent module contains four ingredients:

- ComK activates itself through a transcriptional autoregulatory positive-feedback loop.

- ComK inhibits ComS expression through indirect transcriptional repression.
- MecA inhibits ComK through enzymatic degradation.
- ComS inhibits the latter reaction through competitive binding to MecA.

ComK therefore has two feedback paths.

- The autoregulatory positive-feedback loop:

$$\text{ComK} \rightarrow \text{ComK}$$

- The indirect negative-feedback loop:

$$\text{ComK} \uparrow \dashv \ \text{ComS} \downarrow \dashv \ \text{ComK degradation} \uparrow \dashv \ \text{ComK} \downarrow$$

`Increasing ComK ultimately increases its own degradation.`

The combination of fast positive feedback and slower ComS-mediated negative feedback produces transient competence.

Biological notation

Note that different notations are used for genes, proteins, and promoters:

- *comK* and *comS* denote the **gene**.
- *comG* is an operon containing genes required for DNA uptake.
- ComK, ComS, and MecA denote the corresponding **protein or peptide**.
- P_{comK}, P_{comS}, and P_{comG} denote the corresponding **promoters**.
- P_{comG}-*cfp*, for example, is a fluorescent reporter of P_{comG} activity.

2. From Experimental Observations to a Minimal Model

Differentiation into the competent state affects a small minority of cells in a non-synchronous manner. The authors built strains in which activities of different pairs of promoters within the circuit could be monitored simultaneously in the same cell, revealing interactions between circuit components. Pairwise combinations of the promoters P_{comG}, P_{comK}, and P_{comS} expressing *yfp* or *cfp* were inserted into standard sites within the *B. subtilis* chromosome. Under the imaging conditions, 26 out of 725 cell division events generated competent cells, corresponding to an overall competence frequency of $3.6 \pm 0.7\%$.

2.1 Identifying ComK as the Central Regulator

The authors first simultaneously measured P_{comG} and P_{comK} promoter activities in single competent cells. P_{comG} and P_{comK} are both regulated by ComK, but P_{comK} also has many other important transcriptional inputs. The experiment evaluates the significance of these other transcriptional inputs into P_{comK} during competence.

In this example, a single cell within the *B. subtilis* microcolony becomes competent and consequently its cell division is blocked. As P_{comG} and P_{comK} activities decline during exit from competence, the elongated cell undergoes multiple septation events and returns to the vegetative state. Quantitative time traces of the fluorescence levels show strong correlation between P_{comG} and P_{comK} (Fig. 3b). This behaviour is representative of all 37 measured events in this strain (Fig. 3c).

The high degree of correlation between P_{comG} and P_{comK} suggests that other transcriptional inputs to the ComK promoter do not significantly affect ComK expression during competence.

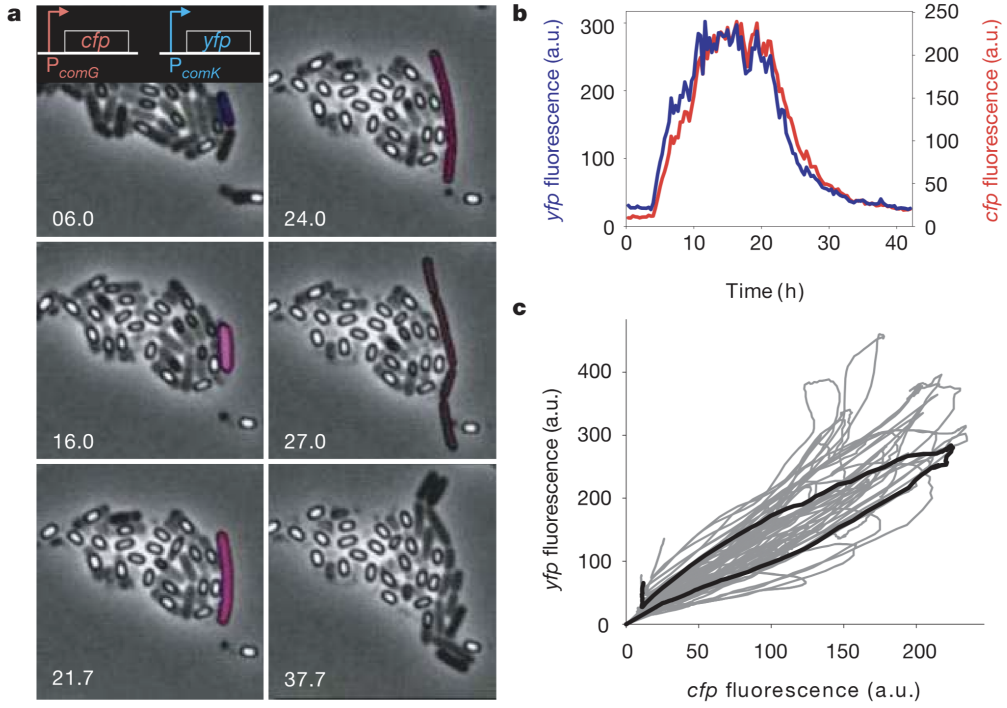


Figure 3. Activities of P_{comG} and P_{comK} promoters are highly correlated during competence. $P_{\mathrm{comK}}-yfp$ and $P_{\mathrm{comG}}-cfp$ activities are coloured blue and red, respectively (see also Supplementary Movie 1). Overlapping dynamics of PcomK and PcomG result in the purple (red plus blue) colour of the competent cell.

2.2 Evidence for ComS-Mediated Negative Feedback

In the MeKS model, ComK indirectly represses ComS, generating an anti-correlation between P_{comG} and P_{comS} activities. Because regulation of *comS* is complex and involves several transcriptional inputs, the authors test the prediction of the model with a strain containing copies of the P_{comG} and P_{comS} promoters expressing *cfp* and *yfp*, respectively.

All cells express P_{comS} to varying degrees. In cells that become competent, P_{comG} activity increases as P_{comS} activity decreases. Later, as P_{comG} activity shuts off and septation begins, P_{comS} activity increases again (Fig. 4). *This negative correlation between* P_{comG} *and* P_{comS} *activity is present during both entry and exit from competence*, although it is more closely synchronized during entry.

This behaviour is representative of all 31 competent cells analysed, and is consistent with negative regulation of comS by ComK. By contrast, the non-competent sister cell does not show a corresponding reduction in P_{comS}, demonstrating that the negative correlation is specific to competence.

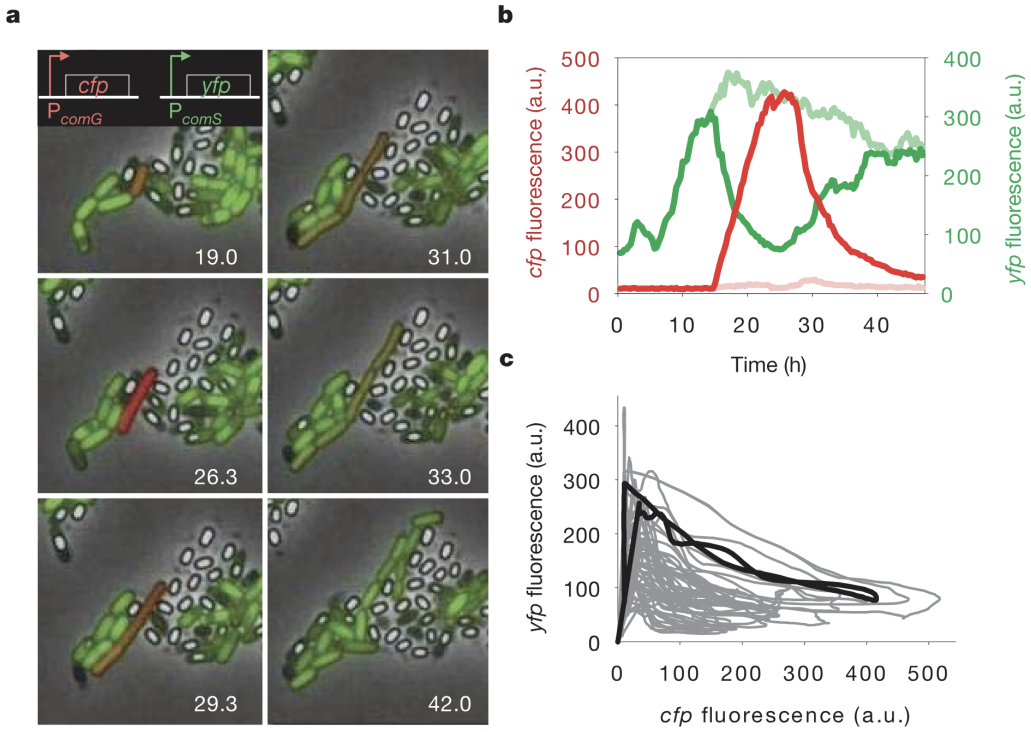


Figure 4. Promoter activities of P_{comS} and P_{comG} are anti-correlated during competence. A typical competence event with $P_{\mathrm{comS}}-yfp$ and $P_{\mathrm{comG}}-cfp$ expression shown in green and red, respectively (see also Supplementary Movie 2).

2.3 Repeated Competence Events and Sister-Cell Variability Support Stochastic Initiation

A fundamental question is whether initiation of competence is stochastic or affected by memory of previous events. Escape from competence returns promoter activities to pre-competence levels, suggesting the possibility of successive episodes of competence. Fig. 5d shows two consecutive competence events can be observed in a single cell lineage, showing that cells retain the potential to re-initiate competence. In fact, re-initiation occurred with a frequency of $6.0 \pm 2.0\%$, not significantly different from the overall competence frequency $3.6 \pm 0.7\%$. Repeated competence events are neither favoured nor suppressed.

This evidence for stochastic initiation of competence is further supported by analysis of competence events in sister cell pairs (Fig. 5e). Cells were not significantly more or less likely to become competent if their sister became competent (conditional frequency $= 4.1 \pm 0.9\%$). When two sisters do become competent together, the amount of time one spends in competence is uncorrelated with that of its sister cell.

These results are consistent with a stochastic and memory-less model for competence initiation and duration.

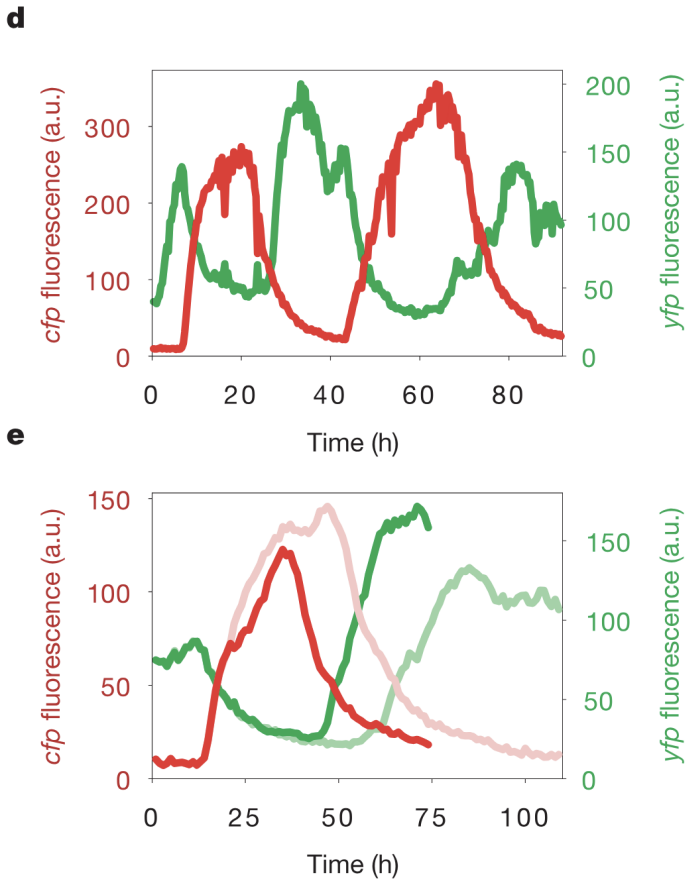


Figure 5. **d**, Consecutive competence events are neither favoured nor suppressed. **e**, The different durations of competence in sister cells.

3. Dynamical Model of the MeKS Module

To understand how the MeKS network structure determines the dynamics of competence, the authors build a mathematical model constrained by experimental observations. This model can be reduced to a system of two stochastic ordinary differential equations incorporating both the direct positive and the ComS-mediated negative feedback loops of ComK.

Here we start from the dimensional rate equations and reduce them to the dimensionless model in equations (S10)–(S11).

3.1 From Molecular Interactions to Rate Equations

ComK inhibits ComS expression through an effective Hill repression function. The enzymatic degradation reactions are assumed to have the standard Michaelis–Menten form:

$$\mathrm{MecA} + \mathrm{ComK} \underset{\gamma_{-a}}{\overset{\gamma_a}{\rightleftharpoons}} \mathrm{MecA{-}ComK} \xrightarrow{\gamma_1} \mathrm{MecA}$$

$$\mathrm{MecA} + \mathrm{ComS} \underset{\gamma_{-b}}{\overset{\gamma_b}{\rightleftharpoons}} \mathrm{MecA{-}ComS} \xrightarrow{\gamma_2} \mathrm{MecA}$$

We use K and S to denote the protein concentrations of ComK and ComS. The concentrations of free MecA, MecA–ComK and MecA–ComS are denoted by M_f, M_K and M_S, respectively. The dynamics of the MeKS module can be described by the following rate equations:

$$\frac{dK}{dt} = \alpha_k + \frac{\beta_k K^n}{k_k^n + K^n} - \gamma_a M_f K + \gamma_{-a} M_K \tag{S1}$$

$$\frac{dS}{dt} = \frac{\beta_s}{1 + (K/k_s)^p} - \gamma_b M_f S + \gamma_{-b} M_S \tag{S2}$$

$$\frac{dM_K}{dt} = -(\gamma_{-a} + \gamma_1) M_K + \gamma_a M_f K \tag{S3}$$

$$\frac{dM_S}{dt} = -(\gamma_{-b} + \gamma_2) M_S + \gamma_b M_f S \tag{S4}$$

The first two terms of equation (S1) represent basal ComK expression and saturating positive autoregulation. The first term of equation (S2) represents ComS expression repressed by ComK. The remaining terms describe reversible binding to MecA. Moreover, we assume that the total amount of MecA is conserved:

$$M_f + M_K + M_S = M_{\mathrm{total}} = \mathrm{constant} \tag{S5}$$

The meaning of the parameters are given in Table 1.

Table 1: Parameters of the dynamical model

Parameter	Description
α_k	Basal expression rate of ComK
β_k	Saturating expression rate of ComK positive feedback
β_s	Unrepressed maximal expression rate of ComS
k_k	ComK concentration for half-maximal ComK activation
k_s	ComK concentration for half-maximal ComS repression
δ_k	Unrepressed degradation rate of ComK
δ_s	Unrepressed degradation rate of ComS
Γ_k	ComK concentration for half-maximal degradation
Γ_s	ComS concentration for half-maximal degradation
n	Hill coefficient of ComK positive feedback
p	Hill coefficient of ComS repression by ComK

3.2 From the Dimensional Model to Two Differential Equations

Assuming that formation and relaxation of the MecA complexes are much faster than the production and degradation of ComK and ComS, thus M_K and M_S may be adiabatically eliminated by setting equations (S3) and (S4) equal to zero. Substitution into the conservation relation (S5) leads to the reduced model:

$$\frac{dK}{dt} = \alpha_k + \frac{\beta_k K^n}{k_k^n + K^n} - \frac{\delta_k K}{1 + K/\Gamma_k + S/\Gamma_s} \tag{S6}$$

$$\frac{dS}{dt} = \frac{\beta_s}{1 + (K/k_s)^p} - \frac{\delta_s S}{1 + K/\Gamma_k + S/\Gamma_s} \tag{S7}$$

where

$$\Gamma_k = \frac{\gamma_{-a} + \gamma_1}{\gamma_a} \qquad \Gamma_s = \frac{\gamma_{-b} + \gamma_2}{\gamma_b} \tag{S8}$$

and

$$\delta_k = \frac{\gamma_1 M_{\text{total}}}{\Gamma_k} \qquad \delta_s = \frac{\gamma_2 M_{\text{total}}}{\Gamma_s} \tag{S9}$$

We further assume that the degradation rates δ_k and δ_s are equal, $\delta_k = \delta_s \equiv \delta$. In that way, rescaling time with these rates and the concentrations of ComK and ComS with Γ_k and Γ_s, respectively, we obtain the dimensionless model:

$$\frac{dK}{dt} = a_k + \frac{b_k K^n}{k_0^n + K^n} - \frac{K}{1 + K + S} \tag{S10}$$

$$\frac{dS}{dt} = \frac{b_s}{1 + (K/k_1)^p} - \frac{S}{1 + K + S} \tag{S11}$$

where now t, K and S are dimensionless variables. The cooperativities of ComK auto-activation and ComS repression are parameterized by the Hill coefficients n and p, respectively. Enzymatic MecA-mediated degradation affects both ComK and ComS; the form of the corresponding nonlinear degradation terms expresses a competitive mechanism, which is the only source of coupling from ComS to ComK, where increasing S reduces the MecA-mediated degradation of ComK.

Reduction of the model to a set of two coupled differential equations allows a phase-plane analysis of the system dynamics.

3.3 Model Parameters

The values chosen for the dimensionless parameters of the model are given in Table 2. The dimensionless parameters are chosen to produce the phase-plane geometry required for excitability, while remaining biologically plausible. These quantities translate into the real biochemical values listed in Table 1, determined by choosing the time unit $\delta_k^{-1} = \delta_s^{-1} = 0.0014s^{-1}$ to fit the experimentally observed excursion times (around 20 hours), and from educated guesses of the concentrations $\Gamma_k = 500nM$ and $\Gamma_s = 50nM$.

Table 2: Parameter values of the dimensionless model

Parameter	Meaning	Value
a_k	Basal expression rate of ComK	0.004
b_k	Saturating expression rate of ComK positive feedback	0.07
b_s	Unrepressed maximal expression rate of ComS	0.82
k_0	ComK concentration for half-maximal ComK activation	0.20
k_1	ComK concentration for half-maximal ComS repression	0.222
n	Hill coefficient of ComK positive feedback	2
p	Hill coefficient of ComS repression by ComK	5

4. Phase Space Analysis

4.1 Nullclines and Fixed Points

The dynamical behaviour of equations (S10)–(S11) can be analysed graphically by plotting their nullclines and vector field in the ComK–ComS phase space as shown in Fig. 6. For the autonomous system $\dot{K} = f(K, S)$, $\dot{S} = g(K, S)$, the **nullclines** are defined by $f(K, S) = 0$ and $g(K, S) = 0$, and their intersections are **fixed points**.

The ComK nullcline can be written explicitly as

$$S = \frac{K}{a_k + b_k K^n/(k_0^n + K^n)} - 1 - K \tag{1}$$

For the ComS nullcline, define $q(K) = b_s/[1 + (K/k_1)^p]$. From $q(K) = S/(1 + K + S)$, we obtain

$$S = \frac{q(K)(1 + K)}{1 - q(K)} \tag{2}$$

Only positive, finite branches are biologically meaningful.

4.2 Local Linear Stability Near a Fixed Point

Suppose $x^* = (K^*, S^*)$ is a fixed point, let $u = x - x^*$ denote a small deviation from the fixed point. The linearized system for the dynamics near the fixed point gives

$$\dot{u} = J(x^*)u \tag{3}$$

where

$$J = \begin{pmatrix} f_K & f_S \\ g_K & g_S \end{pmatrix}_{(K^*, S^*)} \tag{4}$$

is the Jacobian matrix at the fixed point.

The eigenvalues satisfy the characteristic equation

$$\lambda^2 - \tau\lambda + \Delta = 0 \tag{5}$$

With the trace and determinant defined as

$$\tau = \mathrm{tr}(J) \qquad \Delta = \det(J) \tag{6}$$

Then we can classify the type and stability of the fixed points as:

- $\Delta < 0$: saddle point;

- $\Delta > 0$, $\tau^2 - 4\Delta > 0$: real eigenvalues:
  - $\tau < 0$: stable node;
  - $\tau > 0$: unstable node;
- $\Delta > 0$, $\tau^2 - 4\Delta < 0$: complex eigenvalues, spiral-like local motion:
  - $\tau < 0$: stable spiral;
  - $\tau > 0$: unstable spiral;

For reference, the partial derivatives in the Jacobian matrix are

$$f_S = \frac{K}{(1+K+S)^2} \tag{7}$$

$$f_K = b_k \frac{n k_0^n K^{n-1}}{(k_0^n + K^n)^2} - \frac{1+S}{(1+K+S)^2} \tag{8}$$

$$g_S = -\frac{1+K}{(1+K+S)^2} \tag{9}$$

$$g_K = -\frac{b_s p (K/k_1)^{p-1}}{k_1[1+(K/k_1)^p]^2} + \frac{S}{(1+K+S)^2} \tag{10}$$

Local linearization classifies fixed points, but excitability is a global phenomenon: we also need to inspect the invariant manifolds and the long path followed after threshold crossing.

4.3 Phase Plane Portraits

Plotting equation (1) and (2) with parameters in Table 2 produces the nullclines and fixed points. Since b_k regulates the positive-feedback arm, increasing b_k strengthens ComK autoactivation; whereas b_s controls the ComS-mediated protection of ComK, increasing b_s raises ComS production and inhibites ComK degradation. Thus changing b_k and b_s changes the nullcline geometry. For different values of b_k and b_s, we obtain four typical phase-plane nullcline portraits as shown in Fig. 6.

Here are the four typical phase-plane nullcline portraits:

1. **Monostability:** only one fixed point exists, and it is stable. Typical phase-plane nullcline portraits are vegetative state (A) or competent state (B).
2. **Bistability:** three fixed points exist, two of which are stable. The vegetative and the competent regimes coexist, separated by an unstable saddle at intermediate values of ComK. (C)

3. **Excitability:** three fixed points exist, but only the vegetative state is stable. The competent fixed point is an unstable spiral, and the mid-ComK fixed point remains a saddle point. Large enough perturbations will excite the system out of the vegetative state into the competence region, which can be interpreted as the area around the unstable spiral fixed point. (D)

4. **Limit-cycle oscillations:** only one fixed point exists, but it is unstable. The system exhibits limit cycle oscillations between a mid-ComK and a high-ComK level.

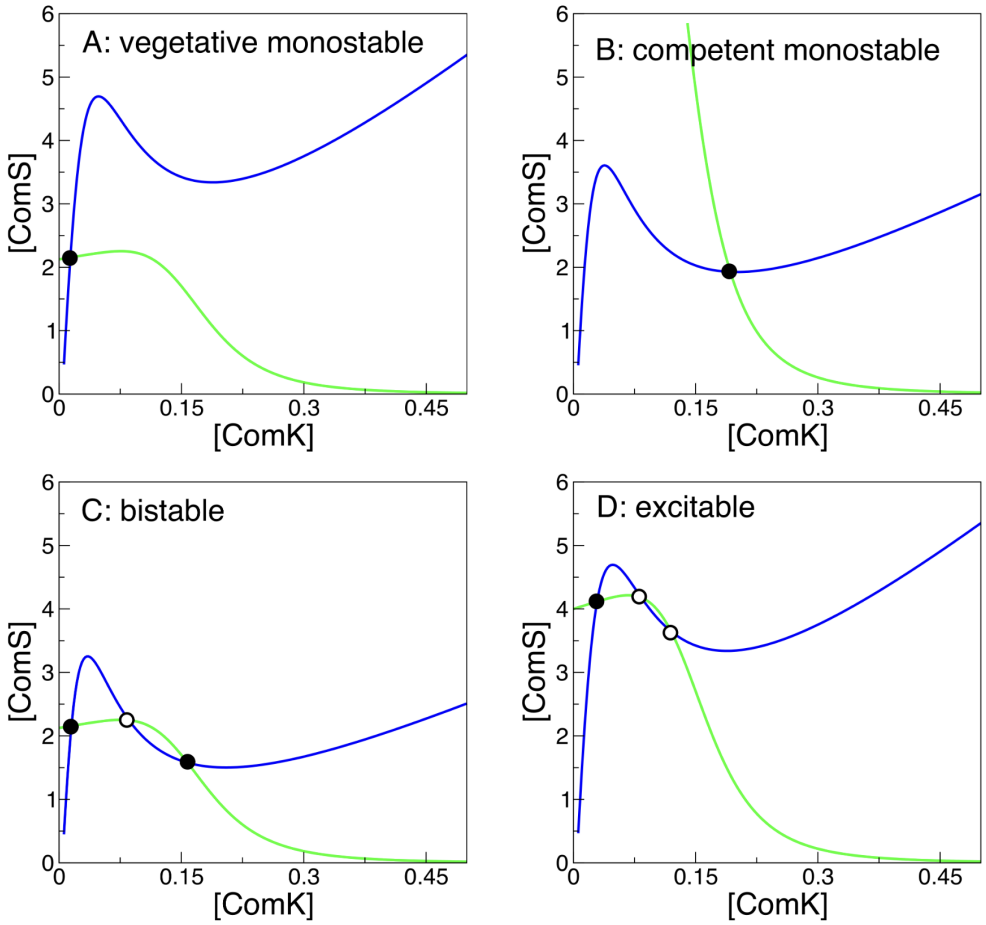


Figure 6. Typical phase-plane nullcline portraits. The blue line corresponds to the ComK nullcline, and the green line to the ComS nullcline. The filled black dots denote stable fixed points, and the empty ones correspond to unstable fixed points. Parameter values are: (A) $b_k = 0.08$, $b_s = 0.68$, (B) $b_k = 0.12$, $b_s = 0.92$, (C) $b_k = 0.14$, $b_s = 0.68$, and (D) $b_k = 0.08$, $b_s = 0.8$.

Transitions Between Phase Portraits

The phase portraits in Fig. 6 represent different dynamical regimes obtained by varying the parameters b_k and b_s. These regimes are connected through bifurcations that change either the number or the stability of fixed points.

From monostability to bistability (A $\rightarrow$ C). Increasing b_k strengthens ComK positive autoregulation and changes the shape of the ComK nullcline. At a saddle-node bifurcation, two additional fixed points emerge: a stable high-ComK state and an intermediate saddle point. The system therefore becomes bistable.

From bistability to competent monostability (C $\rightarrow$ B). Increasing b_s enhances ComS-mediated protection of ComK from degradation. For suitable parameter values, another saddle-node bifurcation eliminates the vegetative stable fixed point and the intermediate saddle, leaving only the stable competent state.

From bistability to excitability (C $\rightarrow$ D). Changes in b_k and b_s can cause the high-ComK fixed point to lose stability through a Hopf bifurcation without changing the number of fixed points. In the excitable regime, the vegetative fixed point is the only attractor, while the high-ComK fixed point is an unstable spiral. Sufficiently large perturbations can then trigger transient excursions through the competence region.

Unlike saddle-node bifurcations, which change the number of fixed points, Hopf bifurcations change their stability and can generate periodic oscillations. The distinction between excitability and sustained oscillation also depends on the presence or absence of a stable limit cycle.

4.4 Phase Diagram: Monostability, Bistability, Excitability, and Oscillation

We further explore the phase diagram of the model in the (b_k, b_s) plane. The phase diagram plots the number of fixed points in color coding, with blue corresponding to one fixed point and red to three. In the blue area below the red region, the only fixed point is the low-ComK vegetative state, whereas in the area above, it is the high-ComK competent state. For the parameters chosen, the competent state undergoes a supercritical Hopf bifurcation (H, purple line). The green lines represent saddle-node (SN) bifurcations.

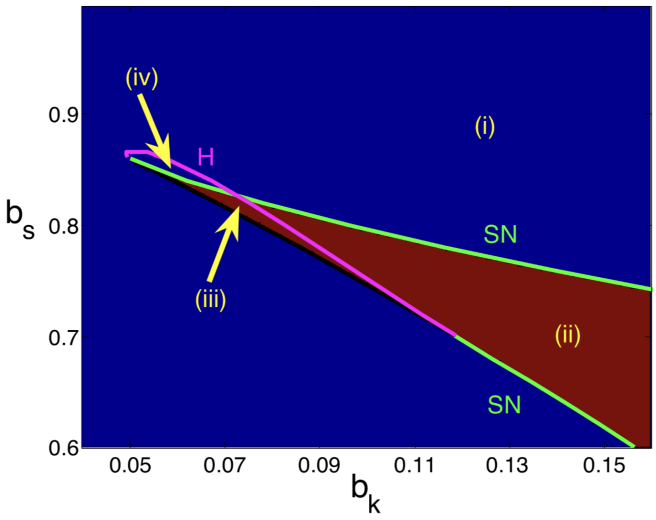


Figure 7. Phase Diagram. Blue represents a single fixed point and red corresponds to three fixed points. The different instability lines delineate four different regions, denoted by yellow numbers (i)-(iv) as described in the previous section.

4.5 Noise-Driven Excitable Dynamics

We now concentrate on the excitable regime of the phase diagram. The dynamical behaviour of equations (S10)–(S11) without noise reveals three fixed points:

1. A **stable node** at low ComK, corresponding to the vegetative state;
2. An **unstable saddle** at intermediate ComK;
3. An **unstable spiral** at high ComK, corresponding to the competence state.

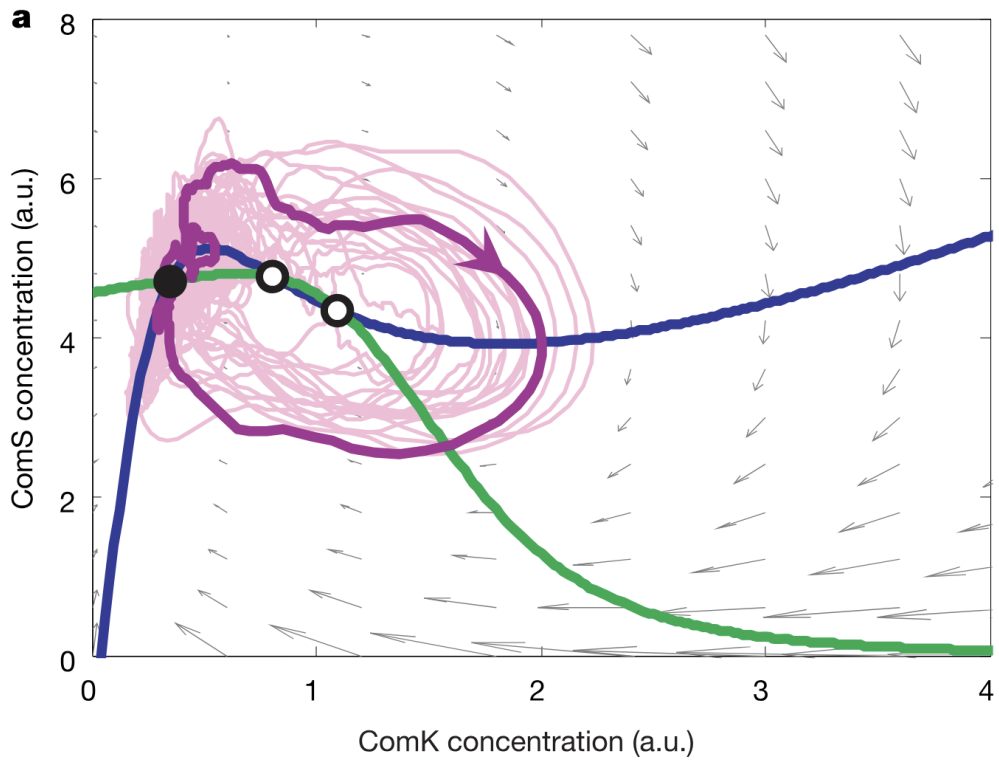


Figure 8. Phase Plane, Nullclines and Fixed Points. Nullclines for ComK and ComS are shown in blue and green, respectively. Grey arrows represent the vector field of the dynamical system. The stable steady-state corresponding to vegetative growth is indicated with a black filled circle. The saddle and the unstable competent fixed points are indicated with open circles. A set of excursion trajectories is shown in pink, with a single representative trajectory of the system highlighted in purple. Initiation of excursions in phase space is triggered by noise, and trajectories are determined by the phase space vector field.

The deterministic nullclines and vector field define the excitable geometry, while noise is required to cross the threshold and initiate the excursions. Stochasticity of gene regulation provides a rich source of such perturbations. Under these conditions, the system is capable of excitable behaviour: relatively small perturbations from the vegetative state may cause long excursions through phase space around the unstable spiral at high ComK.

The intermediate fixed point is a saddle and therefore has one stable and one unstable eigendirection. Its stable manifold forms a **separatrix**: perturbations that remain on the vegetative

side return directly to the stable node, whereas sufficiently large perturbations that cross it generate a large-amplitude excursion through the competence region. The **excitation threshold** is therefore a curve in the two-dimensional (K, S) phase space. The amount of ComS matters because it alters MecA-mediated degradation of ComK.

To understand how the network structure supports the dynamics of competence, we incorporate noise into the differential equation model of the MeKS module. The authors considered three different possible sources of competence-triggering fluctuations:

- Additive noise acting on ComS promoter activity:

$$\frac{dS}{dt} = \frac{b_s}{1 + (K/k_1)^p} - \frac{S}{1 + K + S} + \xi(t) \tag{N1}$$

- Additive noise acting on ComK promoter activity:

$$\frac{dK}{dt} = a_k + \frac{b_k K^n}{k_0^n + K^n} - \frac{K}{1 + K + S} + \xi(t) \tag{N2}$$

- Multiplicative noise acting parametrically on b_s:

$$\frac{dS}{dt} = \frac{b_s + \xi(t)}{1 + (K/k_1)^p} - \frac{S}{1 + K + S} \tag{N3}$$

They found that all three forms of perturbation are able to trigger competence events. Here we consider the first case, random fluctuations in ComS expression. Samples of such trajectories, generated by numerical integration of the model, are superimposed as pink lines in Fig. 8.

4.6 How Does the Circuit Generate Transient Competence Events?

The excitable behaviour is caused by the combination of a fast positive and slow negative feedback. In the present system, noise can induce escape from the otherwise stable vegetative state and turn the system 'on' via the ComK positive feedback. On a slower timescale, this initiates the ComS-mediated negative feedback. Reduction in ComS levels eventually shuts the system back 'off' through an increase in ComK degradation, returning the cell to its vegetative state. In this way, ComS has a dual role in the system: on the one hand it is necessary to initiate competence, by blocking degradation of ComK and allowing positive autoregulation to take effect; on the other hand, repression of ComS is necessary for exit from competence, because reduction in ComS levels favours ComK degradation by MecA.

Once noise pushes the state across the separatrix, the deterministic sequence is:

ComK rises $\uparrow$

$\Downarrow$ positive autoregulation

rapid entry into high ComK

$\Downarrow$ indirect repression

ComS gradually falls $\downarrow$

$\Downarrow$ MecA is free

ComK degradation increases

$\Downarrow$

return to the vegetative state

5. Prediction of the Model: Competence Lock through Feedback Bypass

The excitable model of competence induction makes a strong prediction: interfering with the ComS-mediated negative feedback loop should stabilize the competent state without changing the frequency with which cells become competent.

To test this prediction, the authors constructed a feedback bypass (FeBy) strain of *B. subtilis* containing a single chromosomal copy of P_{comG} driving the expression of *comS* (Fig. 9a). This modification leaves the native competence network intact: the added *comS* gene is expressed only in competent cells. By expressing additional *comS* in this way, the natural negative feedback is rendered ineffective.

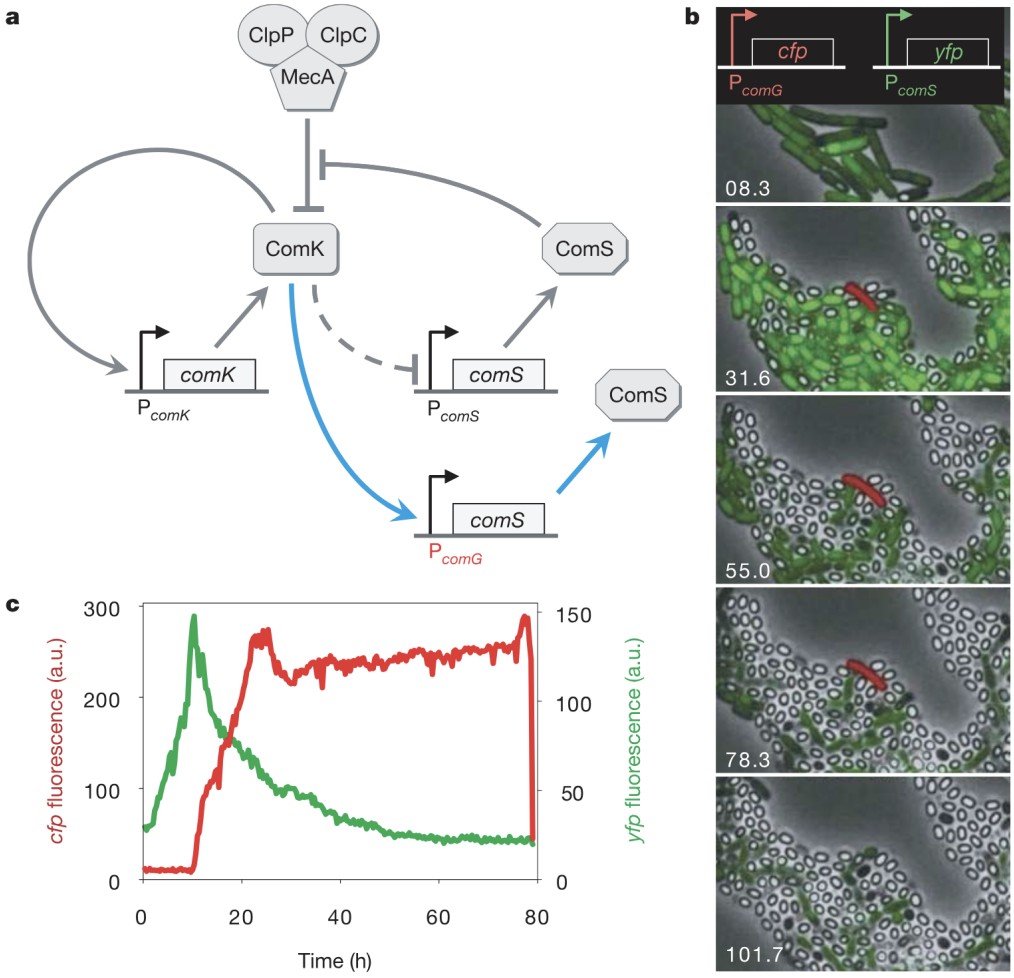


Figure 9. Competence Lock through Feedback Bypass. **a**, Network schematic depicting in blue the extra link introduced to bypass the native ComS-mediated negative feedback loop (FeBy strain). **b**, Frames from film footage of a typical competence event in the FeBy strain, with P_{comG} and P_{comS} activities depicted in green and red, respectively. The FeBy cell enters competence, but cannot exit from competence and eventually lyses (note the sudden drop after

nearly 80 h. See also Supplementary Movie 3). **c**, Quantitative time series show a negative correlation between P_{comG} and P_{comS} in FeBy cells.

The FeBy strain adds a chromosomal $P_{\mathrm{comG}} \rightarrow$comS link while leaving the native circuit intact. This is modeled by introducing an extra term into the ComS equation (S11):

$$\frac{dS}{dt} = \frac{b_s}{1+(K/k_1)^p} + b_g \frac{K^n}{k_g^n + K^n} - \frac{S}{1+K+S} \tag{S12}$$

The additional term represents activation of ComS by ComK via a Hill function with the same cooperativity coefficient as the positive autoregulation of ComK. For $b_g = 0.20$, the trajectories tend to the high-ComK competent fixed point, which has been rendered stable by the feedback bypass link, and whose stability is much larger than that of the vegetative fixed point.

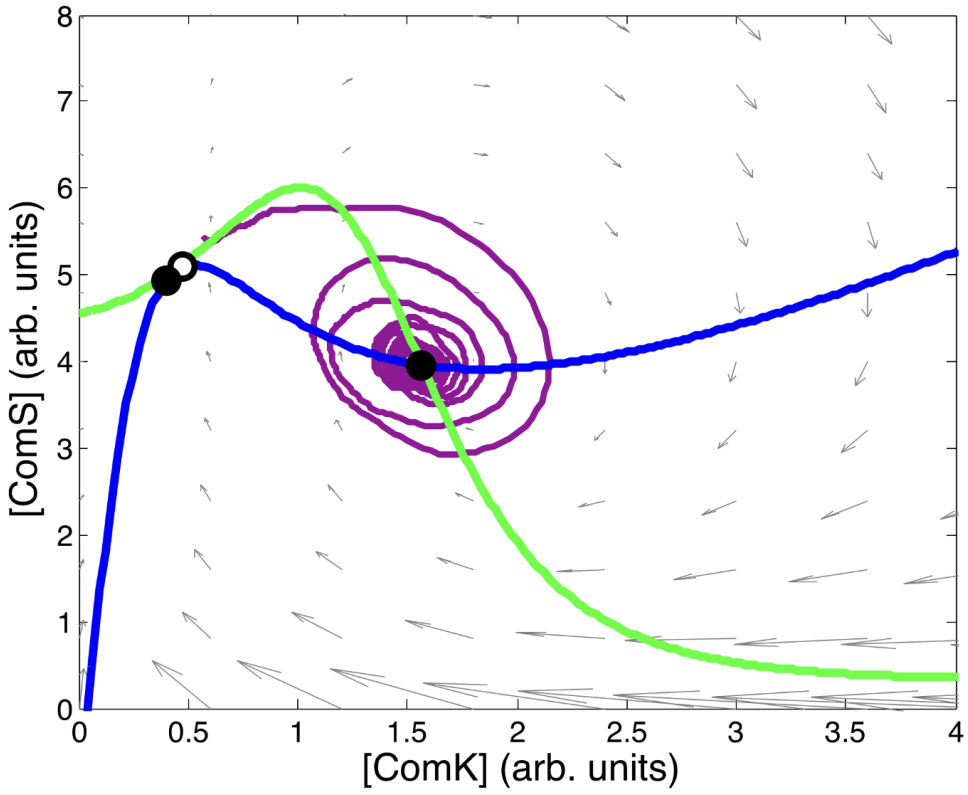


Figure 10. Phase plane portrait of the feedback bypass system. The competent fixed point is now stable. Parameter values are $k_g = k_0 = 0.2$, $b_g = 0.20$.

Time-lapse microscopy on the FeBy strain shows that, *the FeBy strain become competent at a normal frequency*, ($4.2 \pm 0.8\%$, 32 out of 754 events), *but most competent cells fail to return to vegetative growth.* In a wild-type culture, 61% of competent cells successfully exit competence ($n = 136$), compared to only 12% of FeBy cells ($n = 55$). P_{comG} expression levels in competent FeBy cells do not significantly exceed those in wild-type cells, suggesting that the defect in exit is not simply due to overexpression of ComK. As in wild-type cells, a negative correlation between P_{comG} and P_{comS} is observed in FeBy cells (Fig. 9c). Thus, the FeBy cells behave

similarly to wild-type cells during entry into competence, and differ specifically in their reduced ability to exit.

In the FeBy strain, a single rewiring step transformed a transient differentiation process into a terminal one. This reflects a degree of plasticity in the excitable circuit module.

Appendix A. Terms to check after class

1. Buckingham's pi theorem
2. Null cline
3. State space
4. Fixed point
5. Limit cycle

References

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